gotourney

Tournament Director's Guide – running a TGoR Go tournament with the command line tools
Version 0.2 · November 2026

This guide describes the whole tournament, from registration to the EGD upload. All features described here are available in version 0.2.

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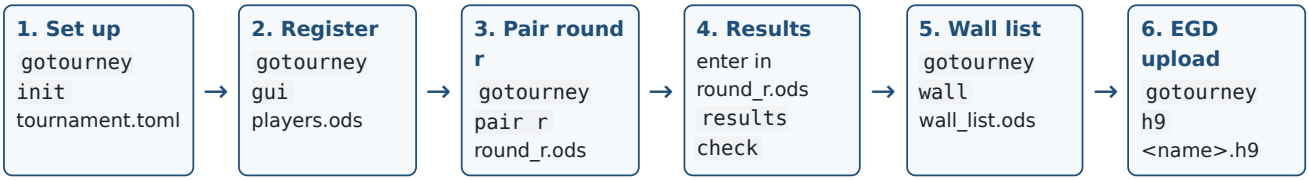
1. The tournament format

Instead of McMahon scores, every player has a **tournament rating (TGoR)**:

- **Start:** the TGoR starts at the player's current EGD GoR on registration day. Players without a GoR start at the value of their declared rank (1k = 2000, 1d = 2100, 100 points per rank). The director may correct a starting GoR, for example when it is clearly outdated. The starting TGoR is capped at the **top bar** (default 3 dan = 2300): stronger players all start at the bar.
- **After every round**, all TGoRs are updated with the EGD rating formula, with two changes. The formula's `con` factor is multiplied by the tournament's **k-factor**, so ratings move faster than in the EGD. The EGD's anti-deflation bonus is not used. All games of a round are rated from the TGoRs *before* that round.
- **Pairing:** players are paired so that the TGoR differences are as small as possible, using the blossom algorithm. Two players never meet twice through automatic pairing, and pairs from the same club are avoided where possible.
- **Handicap:** games are played with **full handicap** by default, one stone per 100 TGoR points of difference. A reduction ("full minus N") can be set. Players may agree on a different handicap for their game.
- **Winner:** the player with the highest final TGoR wins. If two players have the same final TGoR, the one with the *lower* starting GoR ranks higher (the real GoR, not capped at the top bar). If that is also equal, a random draw decides.

2. Overview of the workflow

Each step reads and writes LibreOffice Calc files (`.ods`). You can open, check and change them by hand between the steps. The files are the only record of the tournament: there is no hidden database.



Steps 3 and 4 are repeated for every round.

All files of one tournament live in one folder:

```
my_tournament/
  tournament.toml     settings                                    (step 1)
  players.ods         registered players                           (step 2)
  round_1.ods         pairing of round 1; results are entered here (steps 3, 4)
  round_1.pdf         printable pairing of round 1                 (step 3)
  round_2.ods         ...
  standings.ods       TGoR after each round                        (step 4)
  standings.pdf       printable standings / cross table            (step 4)
  wall_list.ods       final wall list, editable                    (step 5)
  awards.pdf          slides for the award ceremony                (step 5)
  <name>.h9           result file for the EGD                      (step 6)
```

All commands take the option `-d <folder>` to choose the tournament folder. Without it, the current folder is used. So you can either `cd my_tournament` once, or add `-d my_tournament` to every command.

3. Installation AVAILABLE

The tools need Python 3.11 or newer. On Ubuntu also install `python3-venv` and `python3-tk` (the latter for the registration window):

```
sudo apt install python3-venv python3-tk
cd ~/Go/TournamentOrgaTool
python3 -m venv .venv
.venv/bin/pip install -e '.[test]'
source .venv/bin/activate        # afterwards 'gotourney' is available in this terminal
gotourney --help
```

Alternatively, `pipx install -e ~/Go/TournamentOrgaTool` makes `gotourney` available in every terminal without activating anything.

4. Setting up a tournament AVAILABLE

```
gotourney init my_tournament --name "21. Schweriner Drachenschlacht" --rounds 5 --language de
```

This creates the folder with `tournament.toml`, an empty `players.ods` and `translation.toml` (see 4.1). `--language` (`en`, `de` or `fr`, default `en`) sets the language of the PDFs. Open `tournament.toml` in a text editor and fill in the settings **before** registering players:

Setting	Example	Meaning
[tournament] – EGD metadata, written to the header of the .h9 file		
`code`	`""`	EGD tournament code. Leave empty: the EGD assigns it after the upload.
`name`	"21. Schweriner..."	Tournament name (EV).
`place`	"DE, Schwerin"	Country and city (PC).
`dates`	["2026-09-26", "2026-09-27"]	First and last day (DT).
`class`	"A"	EGD tournament class (CL).
`time`	75	Basic thinking time in minutes (TM).
`comment`	"(submitted by ...)"	Comment (CM).
`rounds`	5	Number of rounds (also `init --rounds`).
[output]		
`language`	"de"	Language of the PDFs (pairing, award ceremony): `en`, `de`, `fr` or `custom` (see 4.1).

Setting	Example	Meaning
`translation_file`	"translation.toml"	The texts for `language = "custom"`.
[rating]		
`k_factor`	3.0	Multiplies the EGD `con` factor. Higher values make ratings move faster during the tournament.
`top_bar`	"3d"	Cap for the starting TGoR: a rank (`"3d"` = 2300) or a GoR number (`2350`). `"none"` switches the bar off. Only the start is capped; during the tournament the TGoR can rise above the bar.
[handicap]		
`reduction`	0	"Full minus N" handicap. Written to the .h9 as `HA[h<N>]`: h0 = full handicap, h2 = reduced by 2 stones, h9 = no handicap.
`komi_even`	6.5	Komi in even games: 6.5, 7 or 7.5 (Chinese rules).
`komi_handicap`	0.5	Komi in handicap games.
[pairing]		
`[[pairing.rules]]`	Club, avoid_same, 150	Pairing rules (see 6.3). By default only the club rule is active: pairs from the same club are treated as if their TGoR difference were 150 points larger.
`random_seed`	12345	Seed for random decisions (colours, choosing a joker, final tie-breaks). The same seed always gives the same result.
[registration]		
`gor_outdated_years`	5	Warn if a player's last EGD tournament is older than this.
`rank_gor_tolerance`	300	Warn if declared rank and GoR differ by more than this.
`extra_fields`	["Fee paid", "Age group"]	Additional data you want to record for each player (see 5.6). Each name becomes a column in `players.ods` and `standings.ods` and a field in the registration window.
`[registration.field_values]`	"Fee paid" = ["yes", "no"]	Fixed values for an additional field. Such a field gets a dropdown and only accepts these values (see 5.6).

> Invalid settings are reported with the setting's name, for example `handicap.komi_even must be one of 6.5, 7.0, 7.5`. You can change settings later, including `rounds`. New rounds count as played by all registered players until you change that in `players.ods`. Columns of removed rounds are ignored and disappear the next time `players.ods` is written.

4.1 Language of the PDFs

The printable pairing (`round_r.pdf`) and the award ceremony slides (`awards.pdf`) are written in the language set in `tournament.toml`:

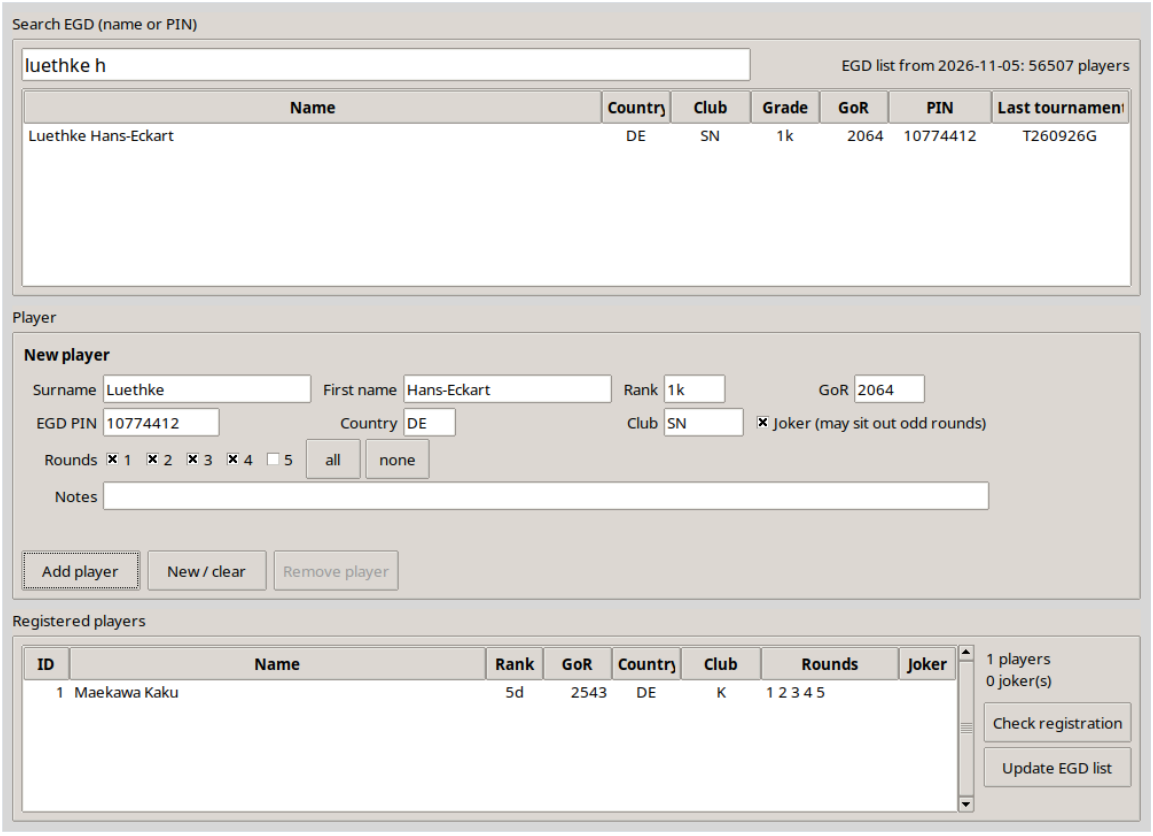
```
[output]
language = "de"                        # en, de, fr or custom
translation_file = "translation.toml"
```

- This covers all fixed texts: column headings, "Round", "Award ceremony", place, games won, the default prize titles, ordinal numbers (1st / 1. / 1re) and the date format (2026-09-26 / 26.09.2026 / 26/09/2026). Prize titles you write yourself in `[awards]` are used as they are.
- The terminal output, the `.ods` files and the `.h9` file stay in English; the `.h9` format is fixed by the EGD.
- **Custom language:** set `language = "custom"` and edit `translation.toml`. `gotourney init` writes this file with the **German** texts as an example; each entry is commented. Keep the placeholders in braces, e.g. `{round}`; `\n` is a line break. Exceptions for ordinal numbers go into the `[ordinals]` section (e.g. `1 = "1re"`). Missing entries are taken from English, with a warning.
- `gotourney translation-file` writes the example file again (`--language en|de|fr` to start from other texts, `--force` to overwrite an existing file).

5. Registering players

5.1 The registration window AVAILABLE

```
gotourney gui -d my_tournament
```



The registration window: EGD search (top), player form (middle), registered players (bottom).

1. **Search:** type part of the name in the search field. Name parts can be in any order ("*erik broek*" finds *van den Broek Erik*) and umlaut spellings are treated alike ("*Lüthke*" finds *Luethke*). Typing digits searches by EGD PIN. Players who are already registered are marked *(registered)*.
2. **Choose a player:** press ↓ and **Enter**, or double-click. **Enter** in the search field takes the first match. The form is filled with name, rank, GoR, PIN, country and club. The window then fetches the player's latest data live from the EGD and updates the GoR if it changed.
3. **Check and complete the form:** untick rounds the player will not play, tick **Joker** if the player may sit out odd rounds, and add notes. Hints in orange appear under the form for:
   - a PIN that is already registered;
   - a rank and GoR that differ by more than `rank_gor_tolerance`;
   - a last EGD tournament older than `gor_outdated_years`, meaning the GoR may be outdated. Correct the GoR field if needed.
4. **Add player** (or Enter in a form field) writes the player to `players.ods` immediately.

- **Players who are not in the EGD:** type their data into the form without searching. A rank is enough: the GoR is then taken from the rank.
- **Editing:** click a player in the bottom table. The form switches to *Edit ID n*. Use **Save changes** or **Remove player**. **New / clear** goes back to adding.
- **Check registration** runs the registration check (see 5.4).
- **Update EGD list** downloads a fresh copy of the EGD player list.

> **The EGD player list.** Searching uses a local copy of the complete EGD player list (about 56,500 players). It is stored in `~/.cache/gotourney/` and downloaded again when it is more than a day old. If there is no internet at the venue, the last downloaded list is used. Run `gotourney egd update` the evening before the tournament to be safe.

5.2 Registering on the command line AVAILABLE

```
gotourney players add --pin 10774412                      # everything else from the EGD
gotourney players add --pin 20633162 --gor 2500 --rounds 1-3
gotourney players add --pin 10774412 -x "Fee paid=yes" -x "Email=hans@example.org"
gotourney players add Doe John --rank 5k --club HH --country DE --joker
gotourney players list
gotourney egd search "luethke h"                          # find a PIN
```

Options of `players add`: `--rank`, `--gor`, `--pin`, `--country`, `--club`, `--rounds` (`all`, `none`, `1,2,4`, `1-3`), `--joker`, `--notes`, and `-x FIELD=VALUE` (repeatable) for additional data. Values you give explicitly win over the EGD data.

5.3 The file players.ods AVAILABLE

Column	Meaning
ID	Running number. It identifies the player in the round files. Never change or reuse it.
Surname, First name	As in the EGD.
Rank	Declared rank (`5k`, `1d`, ...).
GoR	**Starting TGoR.** Filled from the EGD, or from the rank. You may overwrite it.
EGD PIN, Country, Club	EGD data. Club is used for the club penalty in the pairing.
Joker	`x` = may sit out rounds with an odd number of players.
R1 ... Rn	`x` = plays this round. Empty = does not play.
EGD GoR, Last EGD tournament	Information only: the GoR and last tournament in the EGD on registration day.
Additional fields	One column per entry of `extra_fields` (see 5.6).
Notes	Free text.

Players who arrive late, leave early or skip a round: change their round columns. A player who does not play a round keeps their TGoR unchanged.

5.4 Checking the registration AVAILABLE

```
gotourney players check
```

Problems are reported at three levels:

- **Errors** must be fixed before pairing:
  - the same PIN registered twice;
  - a player with neither GoR nor rank;
  - a round with an odd number of players and no joker. Nominate a joker by marking one of that round's players in the Joker column.
- **Warnings** should be checked: rank and GoR far apart, GoR possibly outdated, a player registered for no round, two players with the same name.
- **Info** is for your knowledge only, for example which jokers can sit out in odd rounds, or which players start at the top bar.

The command ends with an error status if there are errors, so it can also be used in scripts.

5.5 Refreshing EGD data AVAILABLE

```
gotourney players sync              # live from the EGD
gotourney players sync --offline    # only the downloaded player list
```

Fetches the current EGD data for all registered players and reports what changed: name, country, club, EGD GoR and last tournament. The declared rank is never changed. The GoR column (starting TGoR) is only filled when it is empty, so your corrections are never overwritten. Once round 1 has been paired, the starting GoRs are frozen.

5.6 Additional player data AVAILABLE

To record more about each player, for example entry fee, age group, e-mail or T-shirt size, list the fields in `tournament.toml`. A new tournament starts with two fields, each with a fixed list of values:

```
[registration]
extra_fields = ["Fee paid", "Age group"]

[registration.field_values]
"Fee paid"  = ["yes", "no"]
"Age group" = ["U7", "U11", "U15", "U19", "Ü50"]
```

- **Dropdown or free text:** a field listed under `[registration.field_values]` gets a dropdown in the registration window and only accepts these values (an empty value is always allowed). A field that is not listed there is free text. To add an e-mail field, for example, add `"Email"` to `extra_fields` and do not give it a value list.
- The registration window shows the fields below *Notes*, and the table of registered players shows them as columns.
- On the command line: `-x "Fee paid=yes"`. Field names and dropdown values are not case-sensitive (`-x "age group=u11"` is stored as `U11`); other values are rejected.
- `players check` warns about values that are not in the list, e.g. typed by hand into `players.ods`.
- `players.ods` gets one column per field, before *Notes*.
- **Columns you add to `players.ods` by hand** are treated the same way: they are kept when the tools rewrite the file, and they appear in the window (after a restart) and in the standings.
- `standings.ods` lists all additional fields after the TGoR column, so you can, for example, sort or filter the standings by age group.
- The values never affect the rating. They affect the pairing only if you write a pairing rule for the field (see 6.3).

6. Pairing a round AVAILABLE

6.1 Workflow

1. **Optional – manual pairings first.** If you want to fix certain games yourself (for example a rematch, or players who must play early), run `gotourney pair 3 --template`. It creates an empty `round_3.ods`. Enter those boards with the two player IDs and mark them `x` in the *Manual* column. The sheet *Participants* lists the players of the round with their TGoR to help you. If you leave **Handicap** empty, the default handicap, komi and colours are filled in (White and Black may be swapped so that the weaker player gets black). If you enter a handicap, your colours and handicap are kept as they are. Manual pairings may be rematches; you get a warning.
2. **Automatic pairing:** `gotourney pair 3` keeps all boards marked `x` in the *Manual* column and pairs all remaining players of round 3. Boards without the mark are paired again, so running `pair` twice gives a fresh pairing.
3. **Check and print.** `pair` also writes `round_3.pdf`, a printable pairing with only the board numbers, the players (colour stone, rank, name), the result columns and the handicap, ready to post or to collect results on paper. Players may agree on a different handicap; just change the Handicap/Komi cells of that board in `round_3.ods`. The tools always use what is written in the file. After editing the pairing by hand, run `gotourney pairing-pdf 3` to update the PDF.

> `gotourney pair` refuses to overwrite a round file that already contains results. Use `--force` only if you really want to pair the round again. A round can only be paired when all results of the previous rounds are entered, and not after a later round exists.

6.2 How the automatic pairing works

- **Participants:** the players with an `x` in this round's column who are not on a manual board.
- **Odd number of players:** one joker sits out, namely the joker with the **most games already played**, chosen at random if tied. If no joker is available, the tool stops and asks you to nominate one.
- **Pair cost:** each possible pair gets the cost `(|TGoR difference| + points of the pairing rules that apply)²` (see 6.3; by default 150 points for two players of the same club). Pairs who already played each other in this tournament, including forfeited games, and pairs forbidden by a rule are excluded.
- **Optimal pairing:** the blossom algorithm (maximum weight matching) chooses the pairing with the smallest total cost. Squaring the cost prefers several medium differences over a single very uneven pair.
- **No pairing possible:** if rematches make a complete pairing impossible (possible with very few players), the tool stops. Pair the problematic players by hand as manual boards and run it again.

6.3 Pairing rules

Pairing rules use the player columns to make some pairs more expensive or impossible. They are set in `tournament.toml`. A new tournament has one active rule, the club rule; the other examples in the file are commented out (off):

```
[[pairing.rules]]
field  = "Club"
type   = "avoid_same"
points = 150

# children (U7, U11, U15) preferably play other children (switched off)
# [[pairing.rules]]
# field  = "Age group"
# type   = "prefer_same"
# points = 200
# values = ["U7", "U11", "U15"]
```

Key	Meaning
`field`	`Club`, `Country`, `Rank` or an additional field (5.6), e.g. `Age group`.
`type = "avoid_same"`	Players with the **same** value: the pair counts as `points` more TGoR difference.
`type = "prefer_same"`	Players with **different** values: the pair counts as `points` more TGoR difference.
`type = "forbid_same"`	Players with the same value are never paired automatically (e.g. a *Family* field).
`type = "require_same"`	Players with different values are never paired automatically.
`values`	Optional. These values form **one group** and count as the same value; all other values together form "the rest". With the example above, children play each other freely (U11 against U15 costs nothing), a child against an adult costs 200 points, and adults among themselves (including Ü50) are not affected.
`groups`	Optional, instead of `values`: several groups, e.g. `[["U7", "U11"], ["U15", "U19"]]`. Values in the same group count as the same; values in different groups, or one in a group and one outside, count as different. A value may only be in one group.

- **Empty cells:** for `avoid_same` and `forbid_same` an empty cell never counts as "the same" (two players without a club are not in the same club). For `prefer_same` and `require_same` two empty cells count as the same value (two adults without an age group).
- **Groups with avoid_same / forbid_same:** the rule then only applies inside a group, and values of one group count as the same. `groups = [["HH", "HHA"]]` on the Club field treats the two clubs as one and

ignores all other clubs.

- Values are compared without regard to case or surrounding spaces.
- **How many points?** The points are added to the TGoR difference, so 200 points means "a child against an adult is as bad as a 200-point difference". A higher value makes the rule stronger, but the pairing will then accept larger rating differences to satisfy it. A simulated tournament help to find a good value.
- To switch a rule off, delete it or put `#` in front of its lines. With no rule at all, the pairing uses only the TGoR difference.
- `gotourney pair` reports how many boards each rule applies to. Manual boards may break rules; you get a warning for forbidden pairs.
- Too many hard rules (`forbid_same`, `require_same`) can make a complete pairing impossible; the tool then stops and asks for manual boards.

6.4 Handicap, komi and colours

TGoR difference	Full handicap	Game
0 – 50	0	Even game, komi `komi_even`, colours balanced
51 – 150	1	Weaker player takes black, no stones, komi `komi_handicap`
151 – 250	2	Weaker player black with 2 stones, komi `komi_handicap`
…	…	…
851 and more	9	Maximum

- The difference is divided by 100 and rounded, with exact halves rounded **down** (150 → 1, 151 → 2).
- With a reduction N, the handicap is `min(full, 9) − N`, but never below 0. In an h2 tournament the default handicap is therefore at most 7. A result of 1 after the reduction is a handicap-1 game, not an even game.
- **Colours in even games** are balanced over the tournament: the player who has had white more often in even games gets black. If both are equal, the colour is drawn at random. Handicap games do not count for the balance.
- **Board order:** board 1 is the pair with the highest TGoR. Manual boards are sorted in as well.

6.5 The round file round_r.ods

Board	White ID	White TGoR	White	White win	Draw	Black win	Black	Black TGoR	Black ID	Handicap	Komi	Manual
1	1	2543	○ 5d Maekawa Kaku	x			● 3d Lewerenz Bernd	2297	3	2	0.5	
2	5	2143	○ 1d Wolff Martin		x		● 1d Moldovan Christian	2158	2	0	6.5	

- The player columns show the colour (○ white, ● black), the registered rank and the name. They are for reading only. The tools identify the players by **White ID** and **Black ID**.
- Black always receives the handicap stones. The TGoR shown is the TGoR before the round.
- The second sheet, *Not playing*, lists the players who sit out the round (joker, not registered).
- The file prints in landscape, scaled to the page width, so a pairing can be printed and posted directly.

7. Entering results and standings

7.1 Results

Enter the result in `round_r.ods` with an **x** in one of the three columns between the player names: **White win**, **Draw** or **Black win**.

White win	Draw	Black win	Meaning	Rating
`x`			White wins	rated
		`x`	Black wins	rated
	`x`		Draw (jigo)	rated, ½ point each
`x!`			White wins by forfeit (`x!` under Black win: Black wins by forfeit)	not rated – as if the game had not taken place (but counts as "played" for the rematch rule)
`!`		`!`	Both lose by forfeit	
			No result yet	–

Upper case `X` and check marks (✓) are accepted as well. Anything else, for example marks in two columns, is reported with the row number.

7.2 Checking results AVAILABLE

```
gotourney results check 3
```

Reports boards without a valid result, players appearing twice, players who are not registered for the round, and (as a warning) rematches.

7.3 Standings AVAILABLE

```
gotourney standings            # all rounds played so far
gotourney standings --round 2  # after round 2
```

Calculates the TGoR of every player after each round and writes `standings.ods`, for checking and for announcing the standings between rounds. Additional player data (5.6) is included as extra columns.

It also writes `standings.pdf` for printing or posting, in the language set in `[output]` (landscape A4): place, name, rank, TGoR with its change since the start, and one column per round showing whom the player played. A round cell such as `○12+` means "white against the player on place 12, won", `●3– (2)` "black against place 3 with 2 handicap stones, lost"; `=` is a jigo, `!` a forfeit, `?` a missing result and `–` a round not played. A legend under the table explains this. Additional fields are not shown in the PDF. TGoRs are calculated with full precision and shown rounded to whole numbers.

8. The wall list and the EGD upload AVAILABLE

8.1 Wall list

```
gotourney wall
```

Calculates the final TGoR and the places, and writes `wall_list.ods`. Places are ordered by final TGoR (rounded), then lower real starting GoR, then a random draw. A second sheet, *TGoR history*, shows every player's TGoR after each round. You can still correct the wall list by hand, for example a name spelling. All results must be entered before the wall list can be made.

8.2 EGD result file

```
gotourney h9                     # writes <tournament code or name>.h9
gotourney h9 -o drachenschlacht.h9
```

Generates the result file in the EGD's `.h9` format from `wall_list.ods`, so your hand corrections are included. Send this file to the EGD. Header (TC stays empty, the EGD fills it in):

```
; TC[]
; CL[A]
; EV[21. Schweriner Drachenschlacht]
; PC[DE, Schwerin]
; DT[2026-09-26,2026-09-27]
; HA[h0]
; KM[6.5]
; TM[75]
; CM[(submitted by ...)]
;.
```

One line per player:

```
 1 Maekawa Kaku          5d DE  K     4  2543  2611  5+/b    6+/w    3+/w2   2+/w    4-/w     |20633162
```

- Place, name (spaces inside a name part become `_`, e.g. `van_den_Broek`), rank, country (`XX` if unknown), club (`XXXX` if none). Names are spelled as in the EGD.
- Wins (jigo = ½), starting GoR (the real GoR, not capped at the top bar), final TGoR. There are no McMahon columns.
- One field per round: `opponent's place`, `+` win / `-` loss / `=` jigo, `/w` or `/b` for the colour, then the handicap (omitted if 0). `0=` means not played: joker round, absent, or forfeit.
- `|PIN` at the end. Players without a PIN have nothing there.

8.3 The award ceremony AVAILABLE

```
gotourney awards                # writes awards.pdf
gotourney awards -o ceremony.pdf
```

Creates a presentation (16:9 PDF, for a projector) with a title slide and one slide per prize winner: the prize, the player's name, rank, club and country, and place, wins and TGoR. Show it in full-screen mode with any PDF viewer. The command also lists the winners in the terminal. All results must be entered first.

The prizes are set in the `[awards]` section of `tournament.toml`. The slides follow the order of the prizes, so you can go from special prizes to 3rd, 2nd and 1st place. A new tournament starts with these prizes:

```
[awards]
title_slide = true

[[awards.prizes]]
type     = "wins"
min_wins = 5                    # the number of rounds

[[awards.prizes]]
type   = "best_in_group"
field  = "Age group"
values = ["U7", "U11", "U15"]
title  = "Best child"           # "Bestes Kind" with --language de

[[awards.prizes]]
type   = "place"
places = [3, 2, 1]
```

type	Who gets the prize	Settings
`place`	The players on these final places, in the given order.	`places = [3, 2, 1]`
`wins`	Everybody with at least this many wins (jigo = ½), in order of place.	`min_wins = 4`
`best_in_group`	The best placed player(s) of a group, e.g. the strongest child. With `groups`, one prize per group.	`field`, `values = [...]` (one group) or `groups = [[...], [...]]`, `count = 1`

type	Who gets the prize	Settings
`rating_gain`	The biggest TGoR gain (final TGoR minus starting TGoR).	`count = 1`

- **title** is the slide heading. Without a title, a default in the output language is used (e.g. "3rd place" / "3. Platz", "5 wins" / "5 Siege"). A title may contain `{place}`, `{ordinal}` (1st, 2nd, ... in the output language), `{wins}`, `{games}`, `{group}` and `{n}` (1, 2, ... within the prize), e.g. `title = "All {games} games won"`.
- **exclusive = true** skips players who already received an earlier prize (earlier in the list), e.g. so that the tournament winner does not also get the prize for the biggest gain.
- A prize that nobody qualifies for (e.g. nobody won all games) is skipped with a note.
- "Best" means best placed in the final standings, i.e. highest final TGoR.

9. How the ratings are calculated AVAILABLE

The TGoR update follows the EGF rating system as revised in 2021. For a game between A and B (Black receives *H* handicap stones):

```
h(H)      = 0 if H = 0, otherwise 100 · (H − 0.5)
            (handicap 1 is worth 50 points, 2 stones 150, 3 stones 250, ...)
β(r)      = −7 · ln(3300 − r − h)
            (h only for the player who receives the stones)
SE_A      = 1 / (1 + exp(β_B − β_A))              expected score of A
con(r)    = ((3300 − r) / 200) ^ 1.6
r_A'      = r_A + k_factor · con(r_A) · (S_A − SE_A)
            S = 1 win, ½ jigo, 0 loss; never below −900;
            the EGD bonus term is not used; komi does not enter the formula
```

Example with `k_factor = 3`. In the EGD (k = 1) the changes would be a third of these.

Game	Expected score of White	Result	Change White	Change Black
White 2000 – Black 1900, even	0.63	White wins	+22.4	−25.2
White 2000 – Black 1900, even	0.63	Black wins	−37.6	+42.3
White 2000 – Black 1750, 2 stones	0.63	Black wins	−37.6	+49.8

The 2 stones are worth 150 points, so the 250-point game in the last row is rated like a 100-point even game. Lower-rated players have a larger `con` and gain or lose more points per game. The formula was checked against the reference values of an independent implementation of the EGF system.

10. Editing files by hand

- Open the `.ods` files in LibreOffice Calc, change what you need, **save in .ods format** and **close the file**.
- Keep the header row. You may add your own columns (they are ignored), reorder columns, or change formatting. Formatting is lost when a tool rewrites the file.
- **Close the file before running a command that writes it.** The tools refuse to write a file that is open in LibreOffice, because one of the two versions would otherwise be lost.
- Never change a player's **ID** once rounds have been paired. Round files refer to players by ID.
- The registration window notices when `players.ods` changes on disk and reloads it within two seconds.
- Error messages name the file and row, for example `players.ods, row 7: invalid rank 'xx'`.

11. Troubleshooting

Message / problem	What to do
`… is open in LibreOffice`	Save and close the file in LibreOffice, then run the command again. If LibreOffice crashed, delete the leftover lock file `.~lock.players.ods#` in the tournament folder.
`tournament.toml not found`	You are not in the tournament folder. Use `-d <folder>` or `cd` into it.
`round r: … players (odd) and no joker`	Mark a player of that round as joker, or change who plays that round.
EGD list not available	No internet and no downloaded list yet. Register players by typing their data, and run `gotourney egd update` when online.
`externally-managed-environment` when installing	Install into the project's virtual environment (`.venv/bin/pip`), see section 3. Do not use `--break-system-packages`.
Wrong result in an earlier round	Correct it in that `round_r.ods`. Standings and wall list are always recalculated from all round files. Rounds that are already paired keep their pairing.

12. Command reference

Command	Purpose	Status
`gotourney init <dir> [--name] [--rounds] [--language]`	Create a tournament folder	AVAILABLE
`gotourney gui [--update-list]`	Registration window with EGD autocomplete	AVAILABLE
`gotourney players add …`	Register a player (with `--pin`: data from the EGD)	AVAILABLE
`gotourney players list`	Show registered players	AVAILABLE
`gotourney players check`	Check the registration	AVAILABLE
`gotourney players sync [--offline]`	Refresh EGD data of registered players	AVAILABLE
`gotourney egd update`	Download the EGD player list	AVAILABLE
`gotourney egd search <query>`	Search the EGD player list	AVAILABLE
`gotourney pair <r> [--template] [--force]`	Pair round r (keeps manual boards)	AVAILABLE
`gotourney pairing-pdf <r>`	Printable pairing → round_r.pdf (also written by `pair`)	AVAILABLE
`gotourney results check <r>`	Check the results of round r	AVAILABLE
`gotourney standings [--round r]`	TGoR after each round → standings.ods and standings.pdf	AVAILABLE
`gotourney wall`	Final wall list → wall_list.ods	AVAILABLE
`gotourney h9 [-o file]`	EGD result file from wall_list.ods	AVAILABLE
`gotourney awards [-o file]`	Award ceremony slides → awards.pdf	AVAILABLE
`gotourney translation-file [--language] [--force]`	Example translation file for `language = "custom"`	AVAILABLE
`gotourney version`	Show the version	AVAILABLE

All commands accept `--help`, for example `gotourney players add --help`.